

LTSpice Notch Quality Report

EEEEEEEA_combined

Auto-detected swept parameters and quality-factor ($Q = f_0 / BW$ @ passband-3 dB) analysis

Run summary

Signal columns analyzed: 1

Total simulation steps : 25

V(act+,act-)

swept parameters: Cp, Itail

Report contents (per signal column):

- 1. All curves — Bode plot (magnitude + phase)
- 2. Magnitude grouped by each swept parameter (one subplot per fixed value)
- 3. Phase grouped by each swept parameter (one subplot per fixed value)
- 4. Quality-factor table for that column

End of report (extra analyses):

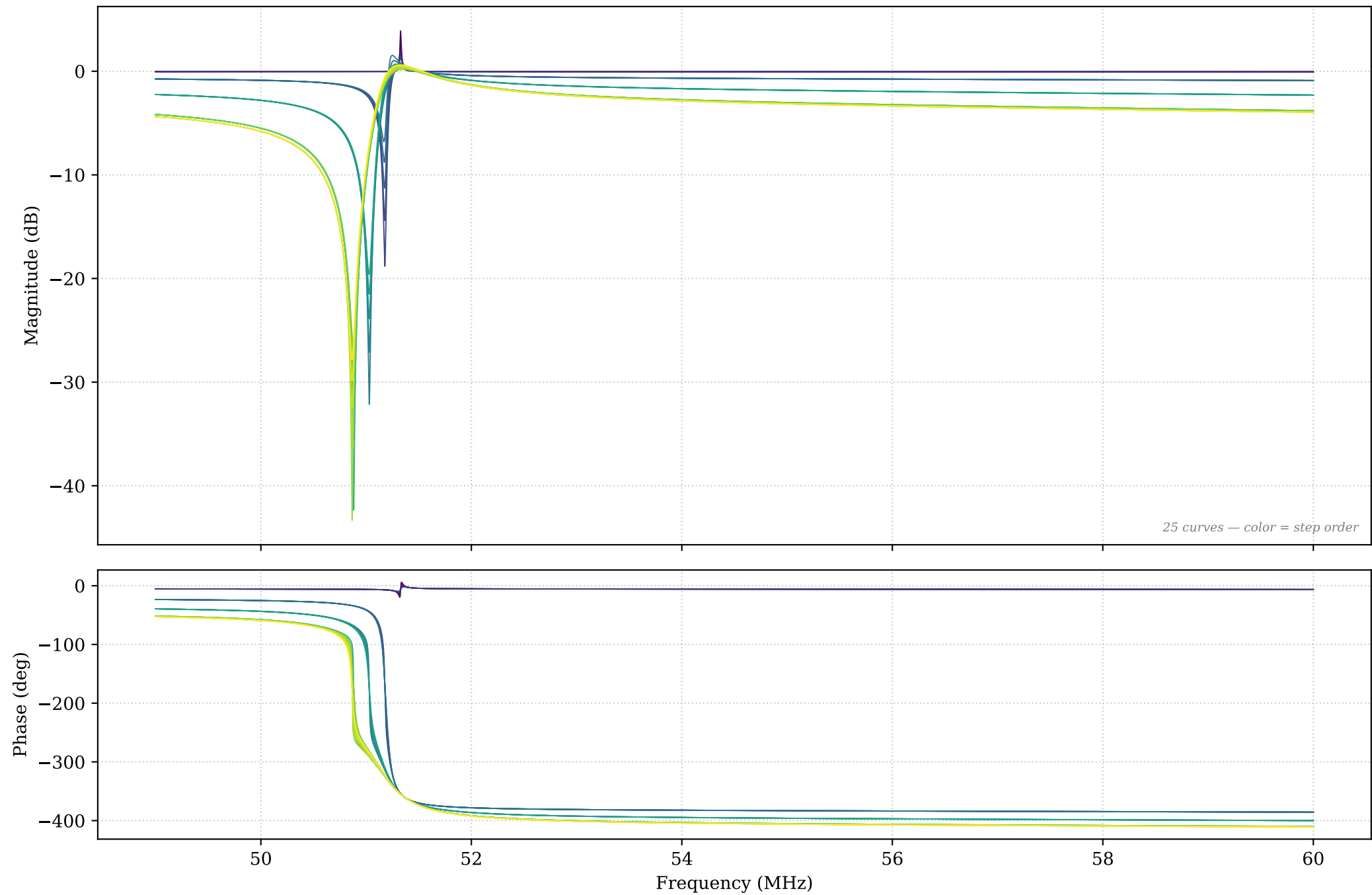
- A. Cross-signal comparison (act vs pas, if both present)
- B. Q vs each swept parameter (one line per other-param combo)
- C. BW vs each swept parameter
- D. Q heatmap for V(act): rows $\times$ columns swept params, colored
- E. Combined quality-factor table across all signals

Signal: $V(\text{act+}, \text{act-})$

Swept parameters: C_p , I_{tail}

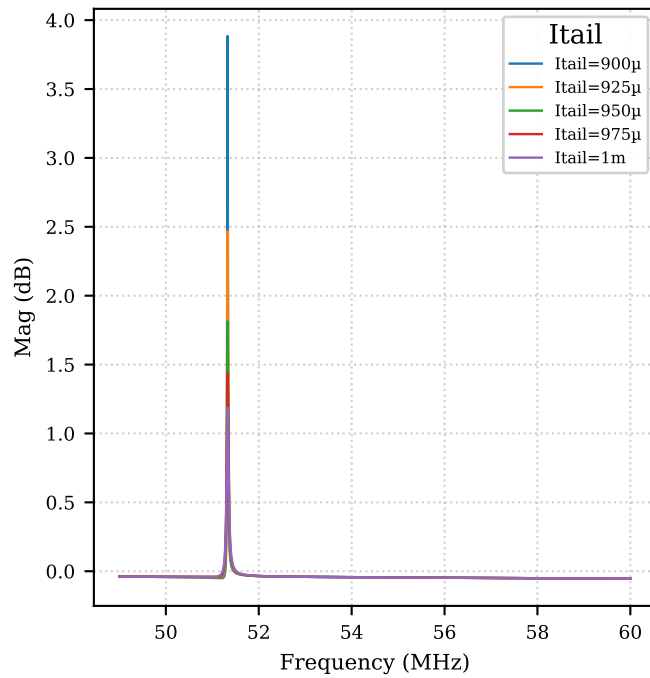
Steps: 25

V(act+,act-) — All curves (25 steps) — Bode

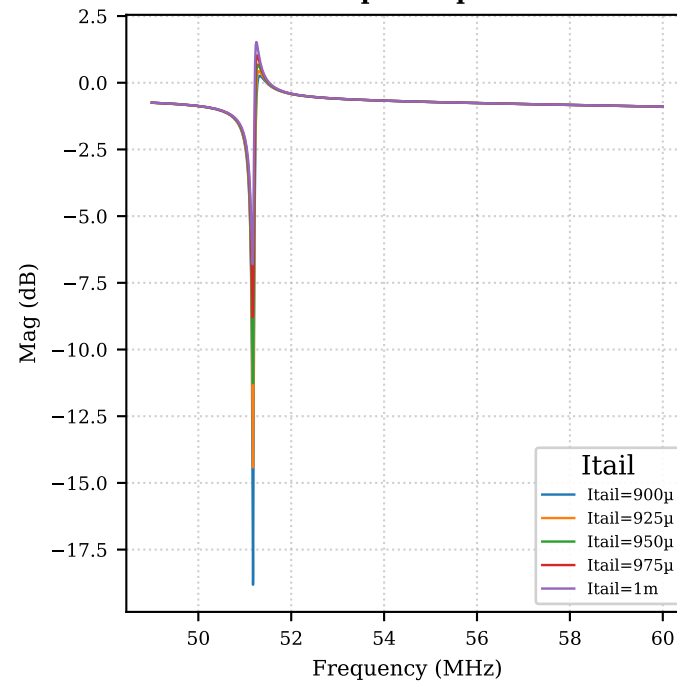


V(act+,act-) — Magnitude grouped by Cp (each subplot fixes Cp; legend = Itail)

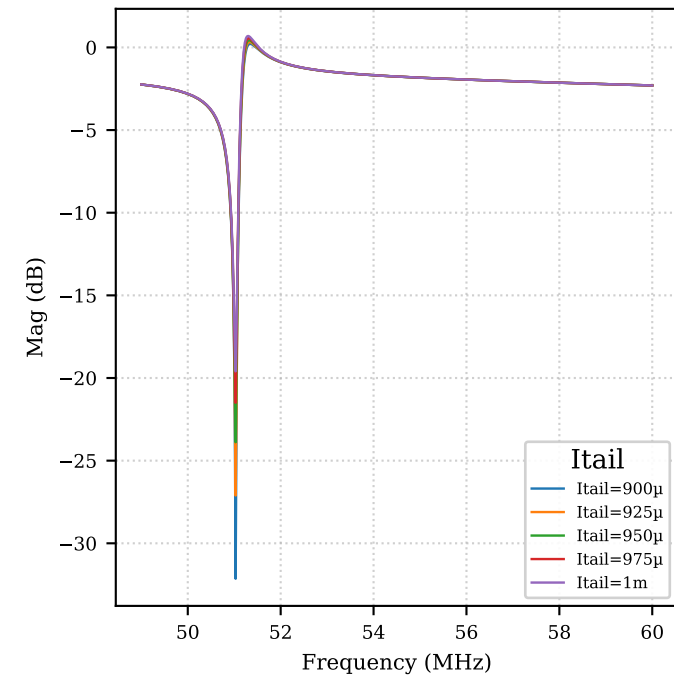
Cp = 100f



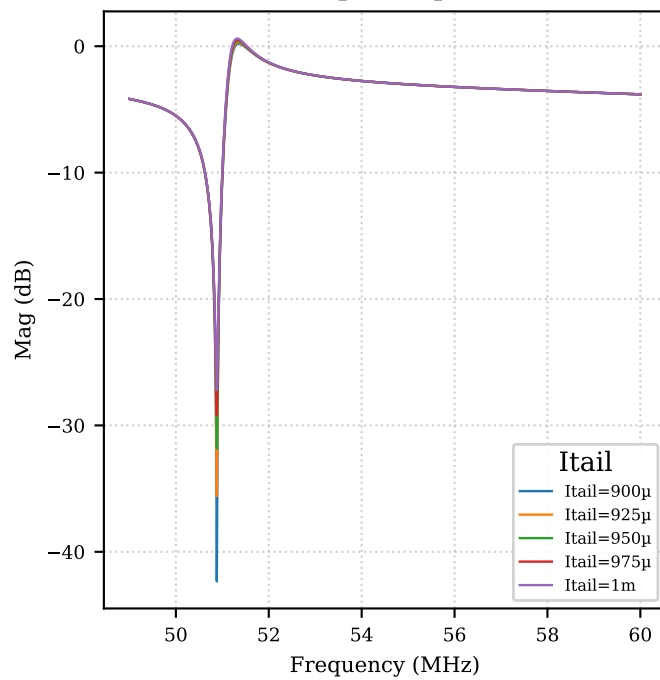
Cp = 1.1p



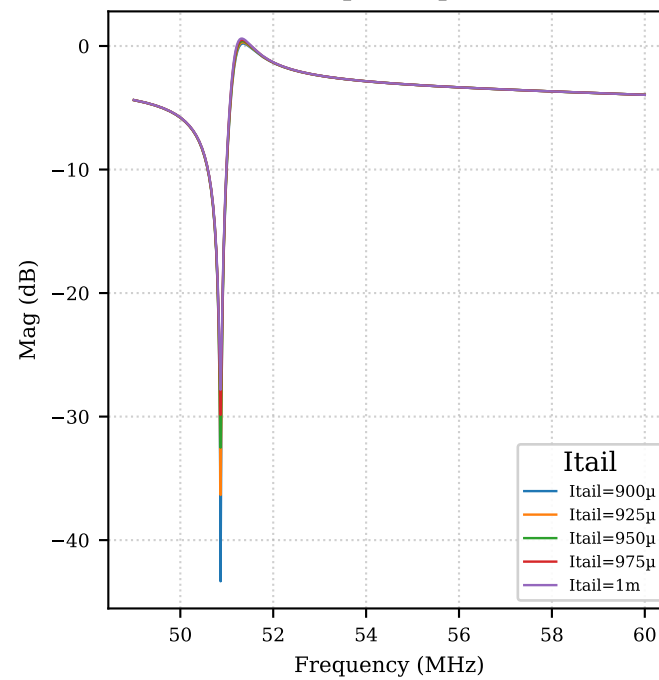
Cp = 2.1p



Cp = 3.1p

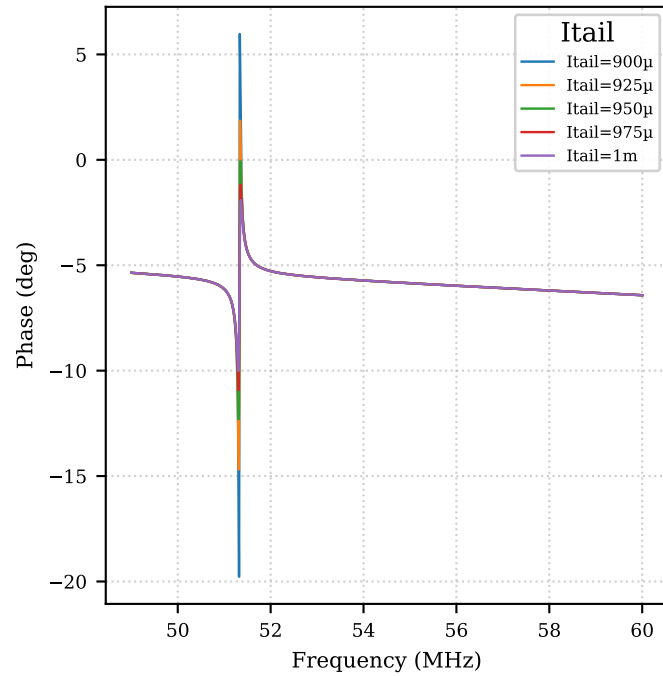


Cp = 3.2p

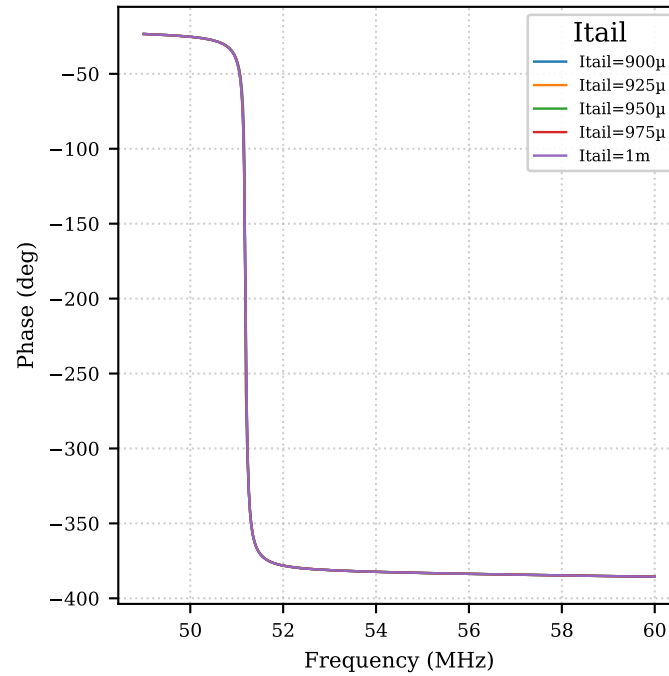


V(act+,act-) — Phase grouped by Cp (each subplot fixes Cp; legend = Itail)

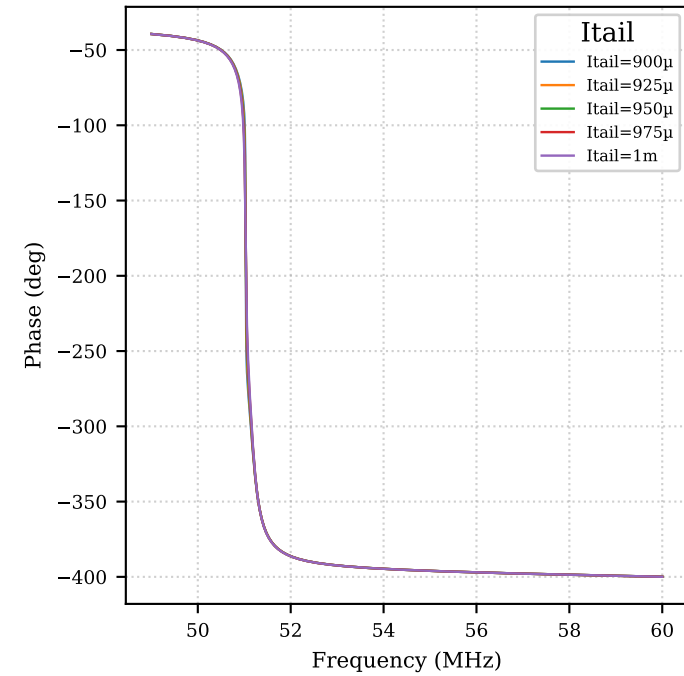
Cp = 100f



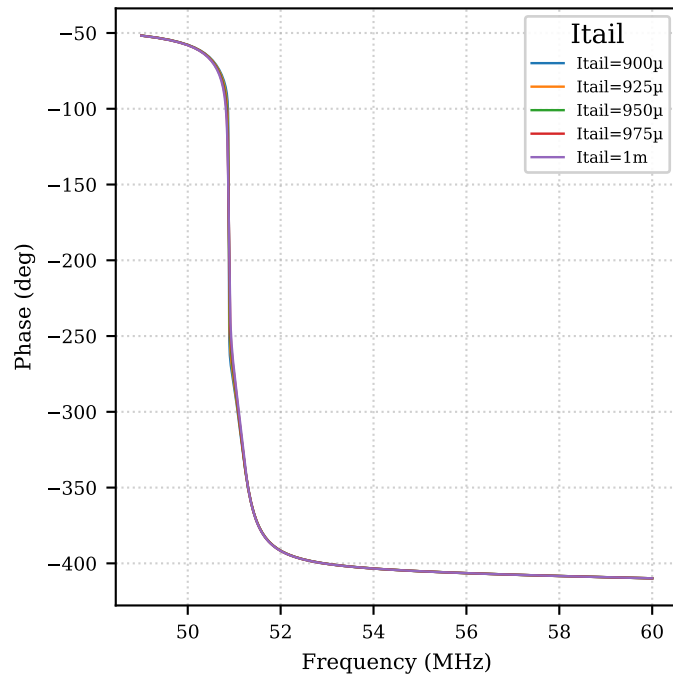
Cp = 1.1p



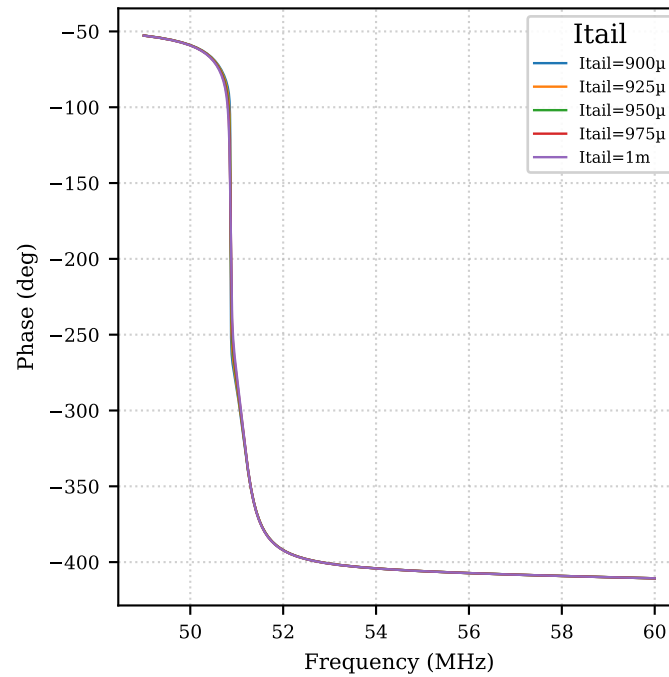
Cp = 2.1p



Cp = 3.1p

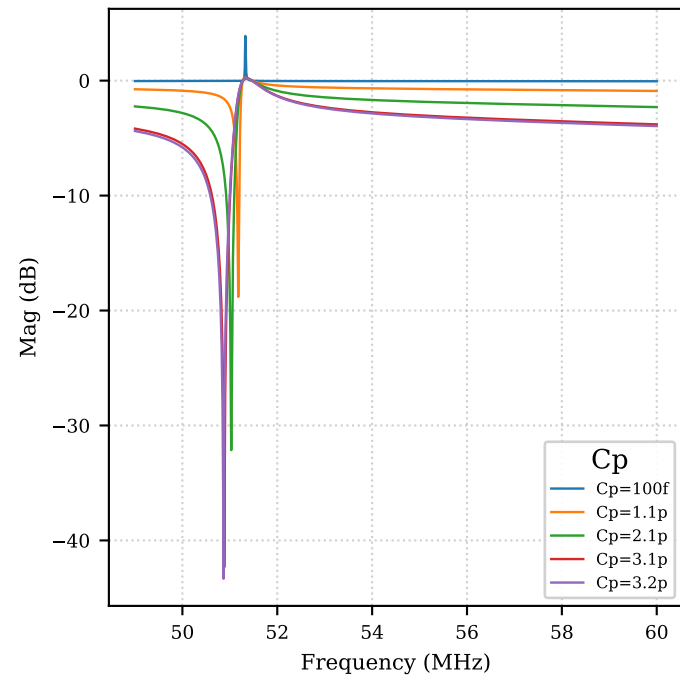


Cp = 3.2p

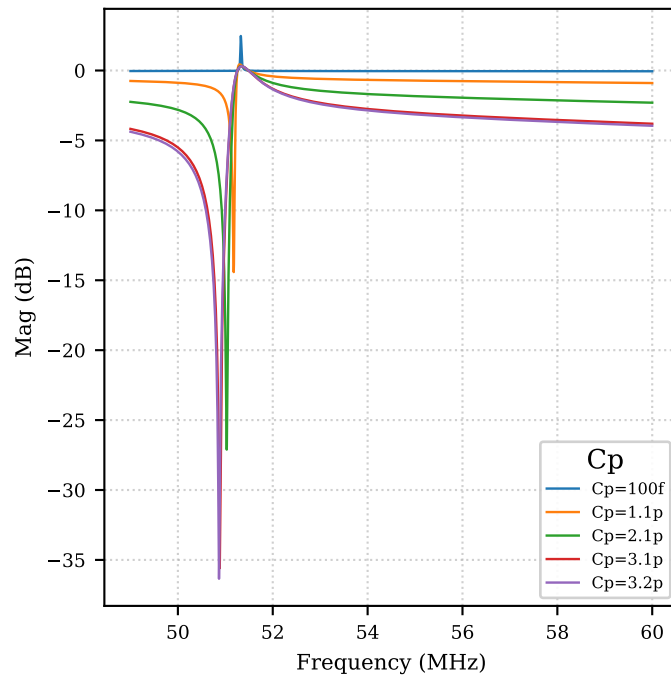


$V(\text{act+}, \text{act-})$ — Magnitude grouped by Itail (each subplot fixes Itail; legend = Cp)

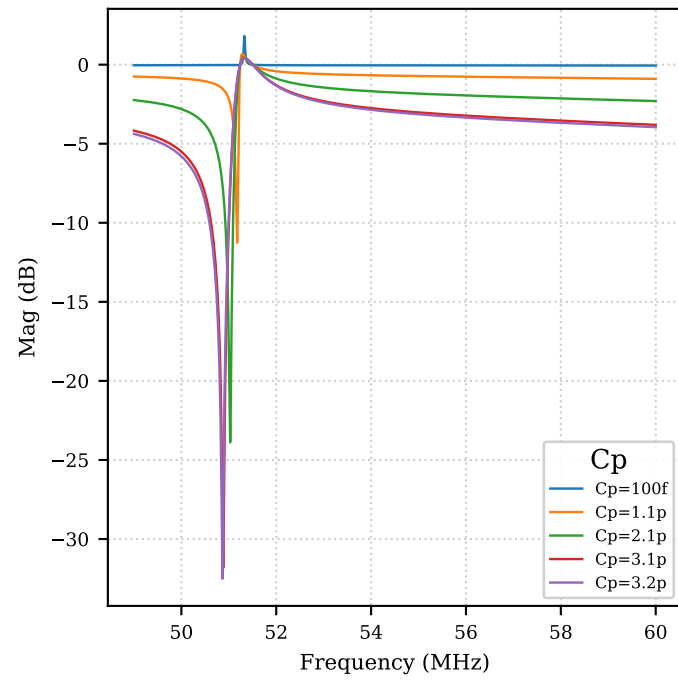
Itail = 900 μ



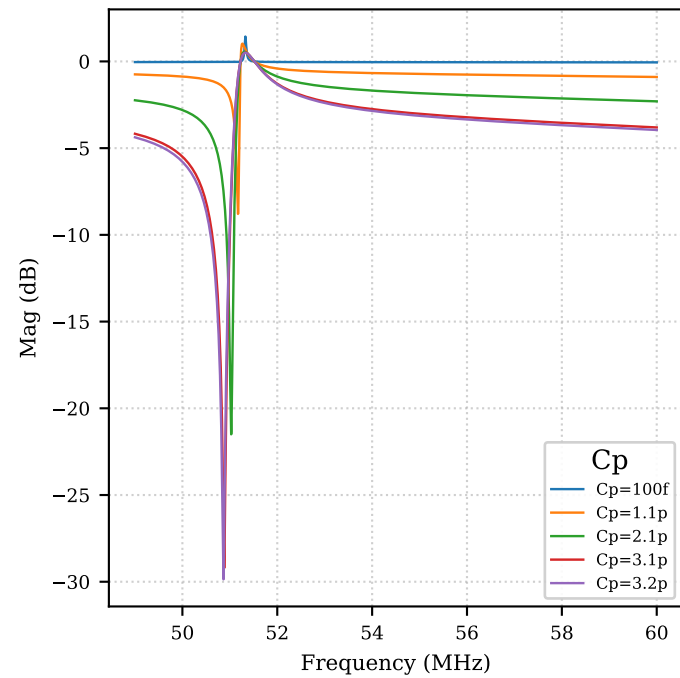
Itail = 925 μ



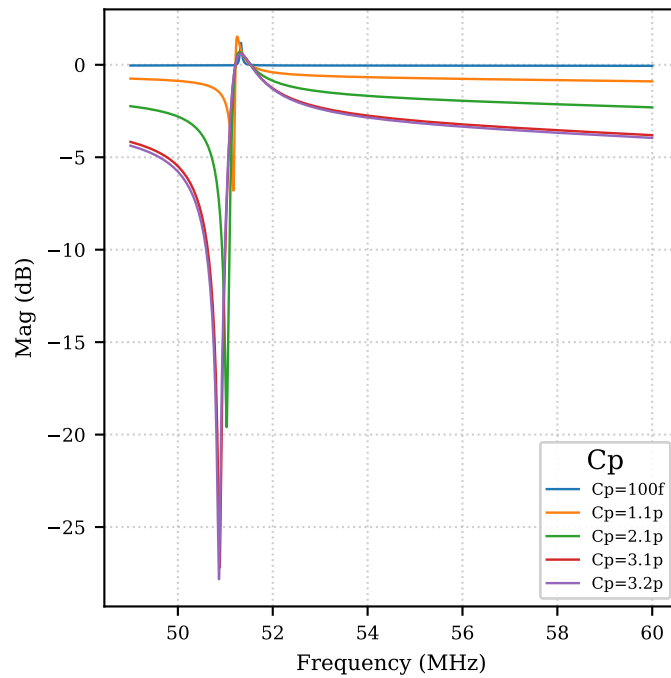
Itail = 950 μ



Itail = 975 μ

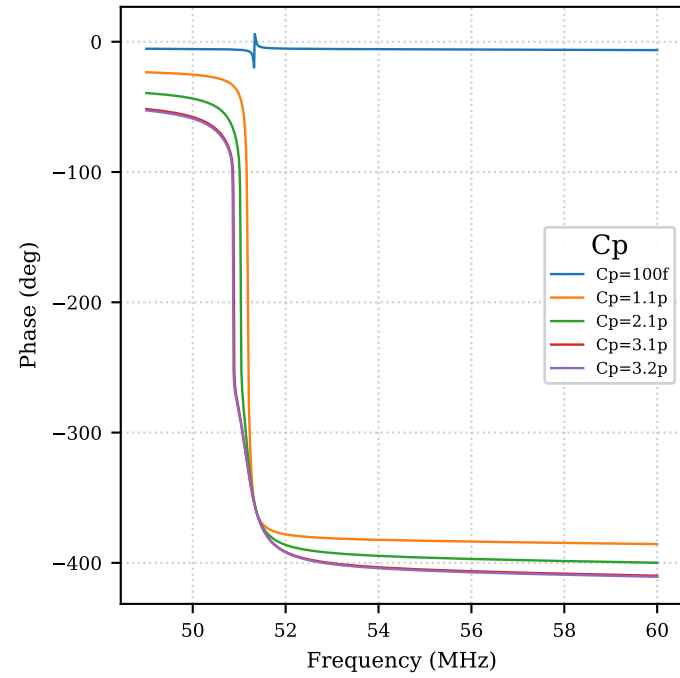


Itail = 1m

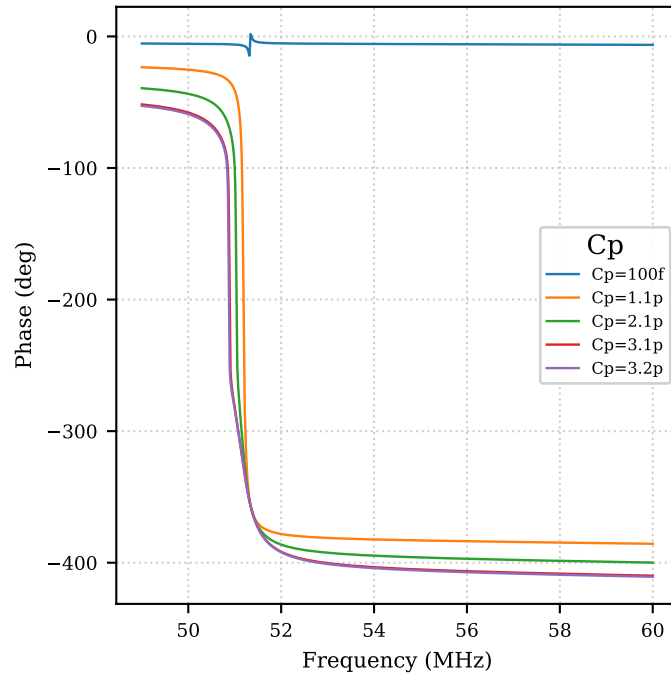


V(act+,act-) — Phase grouped by Itail (each subplot fixes Itail; legend = Cp)

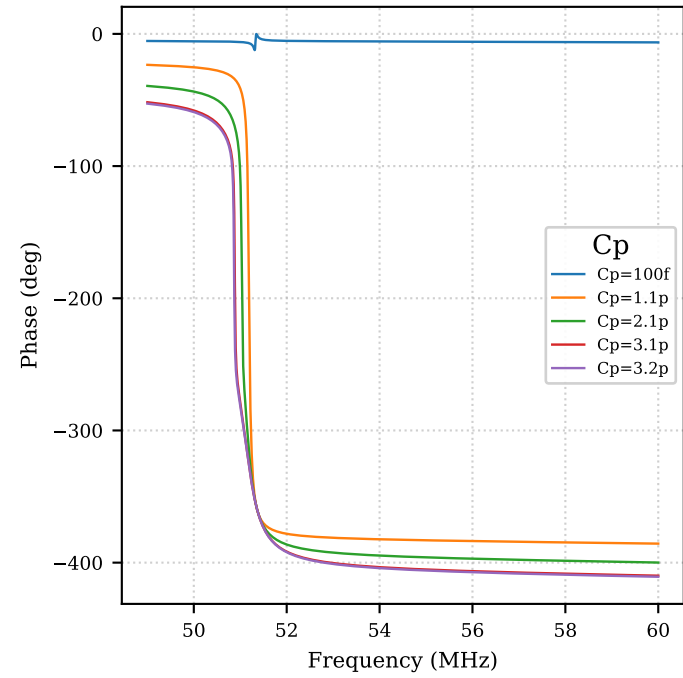
Itail = 900 μ



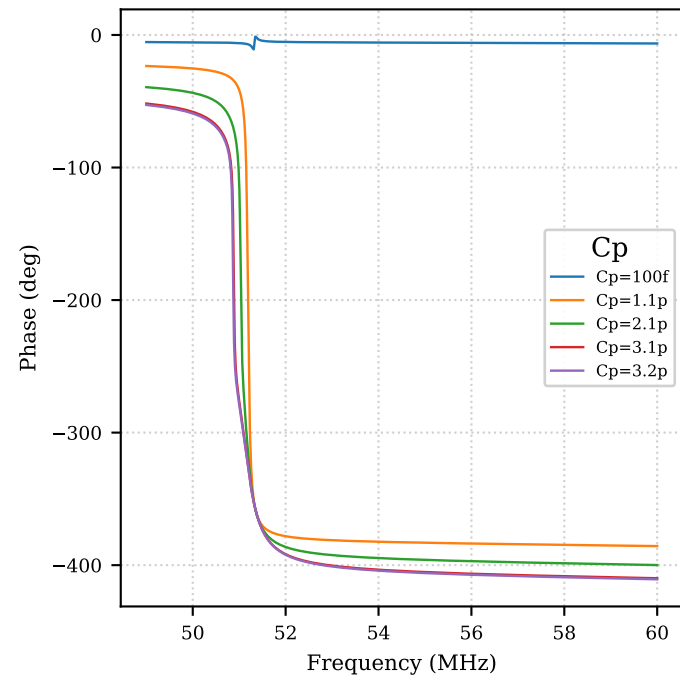
Itail = 925 μ



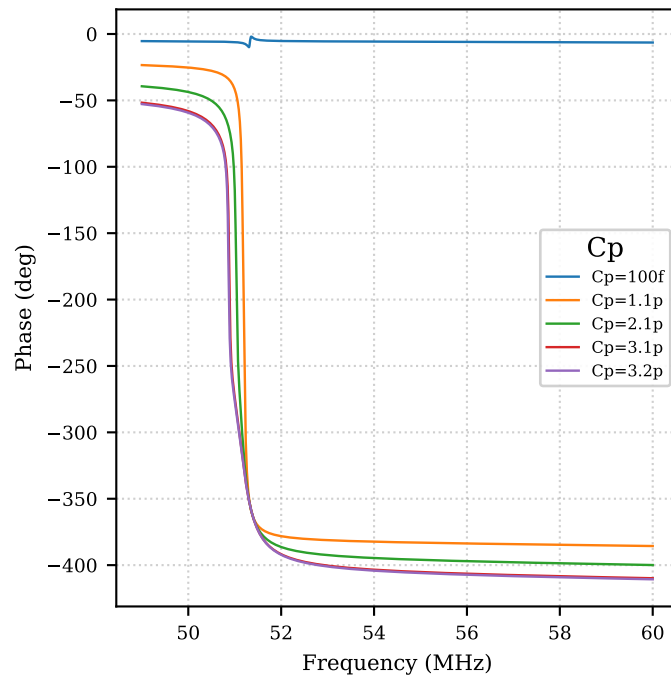
Itail = 950 μ



Itail = 975 μ



Itail = 1m

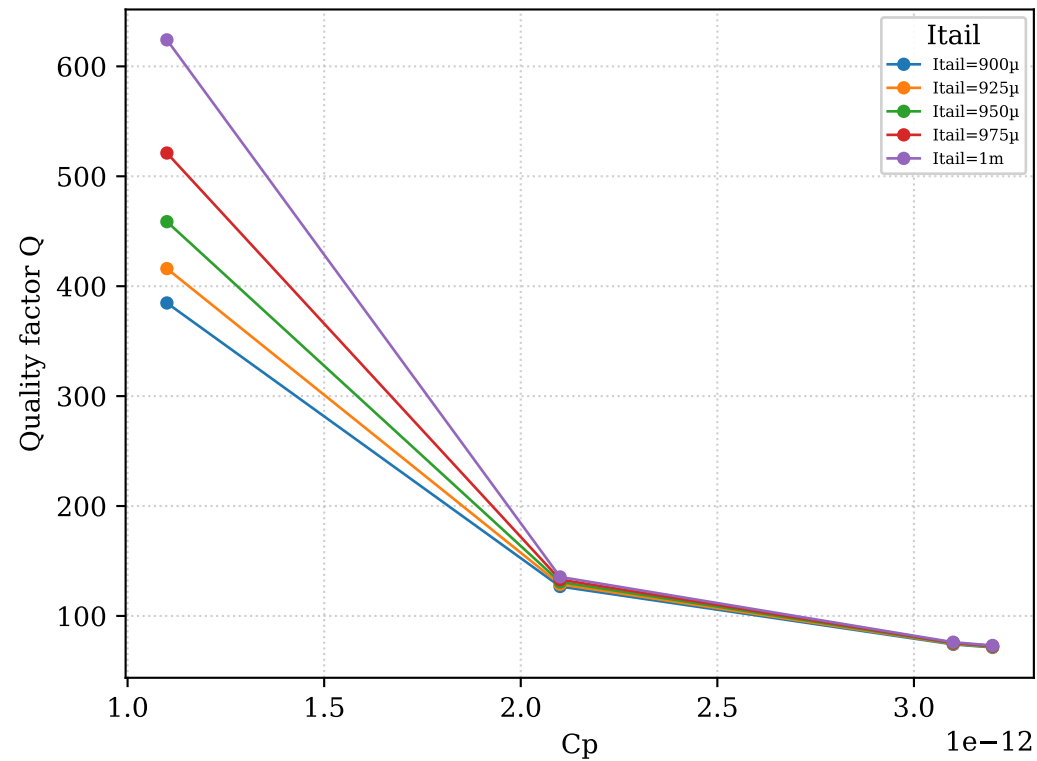


V(act+,act-) — Quality factor table

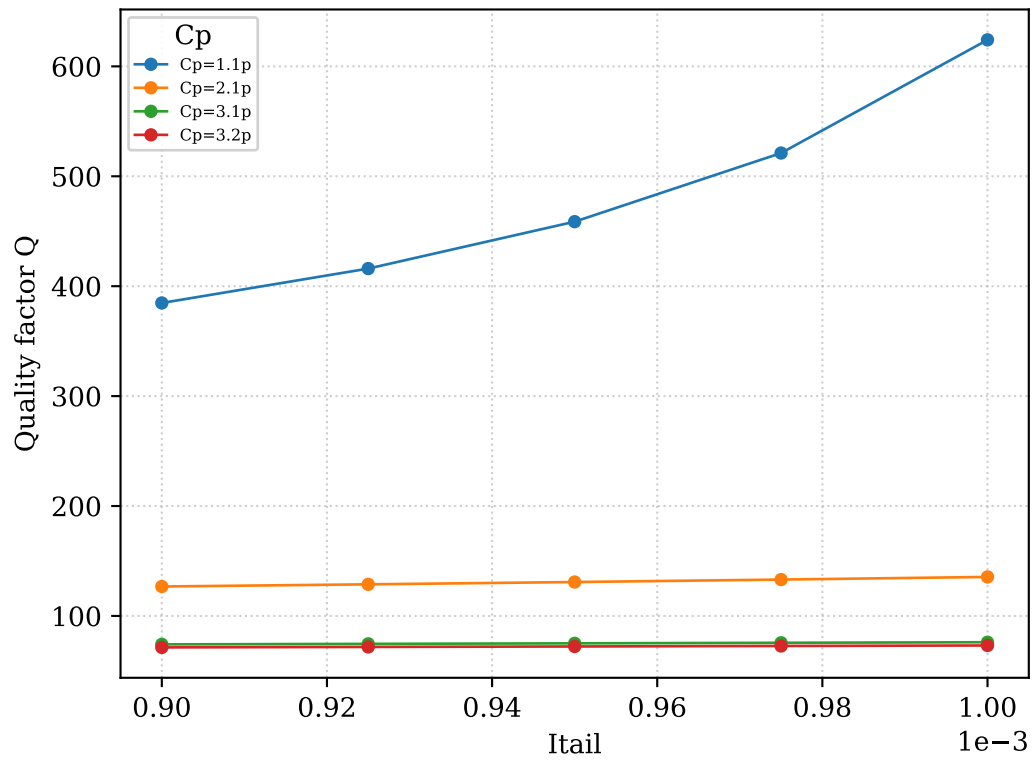
Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	100f	900μ	60	-0.04679	0.007823			notch shallower than 3 dB
V(act+,act-)	100f	925μ	60	-0.04674	0.007866			notch shallower than 3 dB
V(act+,act-)	100f	950μ	60	-0.0467	0.007909			notch shallower than 3 dB
V(act+,act-)	100f	975μ	60	-0.04665	0.007951			notch shallower than 3 dB
V(act+,act-)	100f	1m	60	-0.04661	0.007993			notch shallower than 3 dB
V(act+,act-)	1.1p	900μ	51.18	-0.8662	17.95	1.33e+05	384.7	
V(act+,act-)	1.1p	925μ	51.18	-0.8659	13.55	1.23e+05	416	
V(act+,act-)	1.1p	950μ	51.18	-0.8656	10.4	1.116e+05	458.7	
V(act+,act-)	1.1p	975μ	51.17	-0.8653	7.93	9.819e+04	521.1	
V(act+,act-)	1.1p	1m	51.17	-0.8651	5.938	8.199e+04	624.1	
V(act+,act-)	2.1p	900μ	51.03	-2.291	29.85	4.025e+05	126.8	
V(act+,act-)	2.1p	925μ	51.03	-2.291	24.82	3.963e+05	128.8	
V(act+,act-)	2.1p	950μ	51.03	-2.29	21.6	3.9e+05	130.8	
V(act+,act-)	2.1p	975μ	51.03	-2.29	19.21	3.834e+05	133.1	
V(act+,act-)	2.1p	1m	51.03	-2.29	17.31	3.766e+05	135.5	
V(act+,act-)	3.1p	900μ	50.88	-3.992	38.34	6.864e+05	74.13	
V(act+,act-)	3.1p	925μ	50.88	-3.991	31.6	6.821e+05	74.6	
V(act+,act-)	3.1p	950μ	50.88	-3.99	27.84	6.777e+05	75.08	
V(act+,act-)	3.1p	975μ	50.88	-3.989	25.22	6.732e+05	75.58	
V(act+,act-)	3.1p	1m	50.88	-3.988	23.21	6.687e+05	76.09	
V(act+,act-)	3.2p	900μ	50.87	-4.167	39.17	7.127e+05	71.37	
V(act+,act-)	3.2p	925μ	50.87	-4.166	32.18	7.085e+05	71.79	
V(act+,act-)	3.2p	950μ	50.87	-4.165	28.35	7.043e+05	72.23	
V(act+,act-)	3.2p	975μ	50.87	-4.164	25.7	7e+05	72.67	
V(act+,act-)	3.2p	1m	50.87	-4.163	23.66	6.956e+05	73.13	

B. V(act+,act-) — Quality factor vs swept parameters

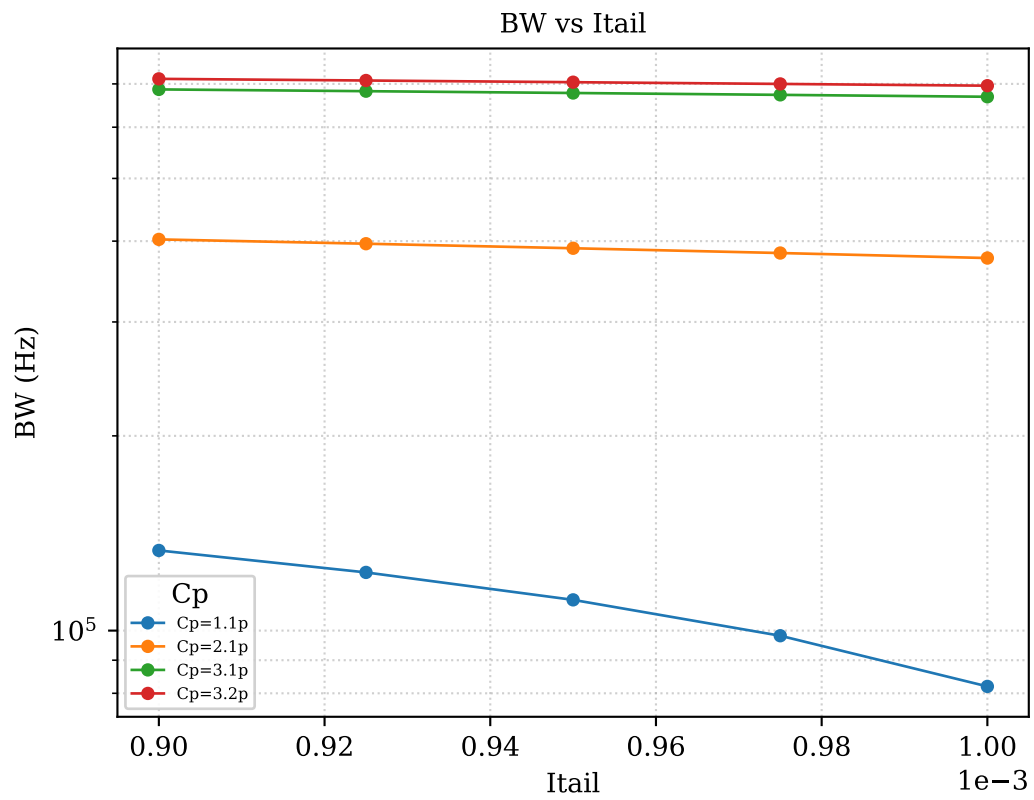
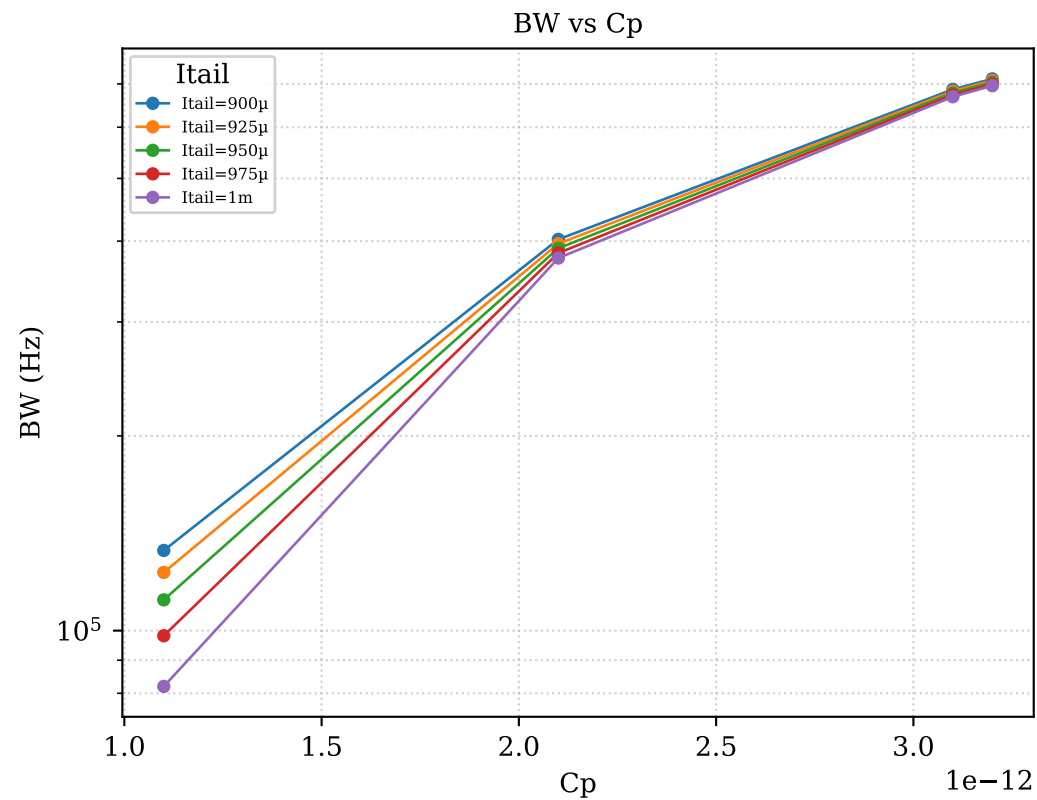
Q vs C_p



Q vs $Itail$



C. V(act+,act-) — Bandwidth (-3 dB) vs swept parameters



D. V(act+,act-) — Q heatmap (rows = Cp, columns = Itail)

	Itail=900μ	Itail=925μ	Itail=950μ	Itail=975μ	Itail=1m
Cp=100f	no BW	no BW	no BW	no BW	no BW
Cp=1.1p	385	416	459	521	624
Cp=2.1p	127	129	131	133	135
Cp=3.1p	74	75	75	76	76
Cp=3.2p	71	72	72	73	73

Red = BW could not be measured | Yellow = Q computed ($Q \leq 1400$) | Green = $Q > 1400$

E. Combined quality-factor table — all signals

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	100f	900μ	60	-0.04679	0.007823			notch shallower than 3 d
V(act+,act-)	100f	925μ	60	-0.04674	0.007866			notch shallower than 3 d
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V(act+,act-)	100f	975μ	60	-0.04665	0.007951			notch shallower than 3 d
V(act+,act-)	100f	1m	60	-0.04661	0.007993			notch shallower than 3 d
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V(act+,act-)	1.1p	950μ	51.18	-0.8656	10.4	1.116e+05	458.7	
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V(act+,act-)	1.1p	1m	51.17	-0.8651	5.938	8.199e+04	624.1	
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V(act+,act-)	3.2p	950μ	50.87	-4.165	28.35	7.043e+05	72.23	
V(act+,act-)	3.2p	975μ	50.87	-4.164	25.7	7e+05	72.67	
V(act+,act-)	3.2p	1m	50.87	-4.163	23.66	6.956e+05	73.13	